

SARKARI SAHAYAK

*Multilingual Document-Aware Agentic Welfare RAG System
Technical Architecture, Design Documentation, and Security Manual*

Prepared for: Prototype Screening Committee

Version: 1.1.0

Status: Showcase Ready

1. Executive Summary & Vision Statement

Accessing government benefits in developing nations remains plagued by structural barriers. Language barriers prevent rural citizens from finding schemes, complex eligibility criteria require reading fine-print regulations, and document illiteracy keeps millions from completing complex registration forms.

Sarkari Sahayak resolves this by serving as an intelligent, conversational agent running on top of WhatsApp - the interface users are already familiar with. By allowing users to talk in their local dialect, upload identity certificates for parsing, and receive pre-filled application forms, the system acts as an agentic counselor, democratizing welfare accessibility.

2. High-Level Human Analogy

Think of the system as walking into a local Welfare Helpdesk office manned by six specialized staff members working in perfect coordination:

- The Receptionist (FastAPI Webhook): Greets you, translates your voice notes, and directs files to the desk.
- The Document Reader (Llama 4 Vision OCR): Glances at your Aadhaar or Income papers, writes key parameters onto a temporary scratchpad, and locks them in a drawer.
- The Librarian (RAG Vector Store): Runs into the records library to find scheme manuals matching your general query.
- The Legal Evaluator (Rules Engine): Cross-checks the details on your scratchpad against the scheme eligibility criteria.
- The Senior Counsel (Llama 3.3 Graph): Suggests complementary schemes and writes a friendly, spoken response in your preferred language.
- The Clerk (FormFiller Service): Fills out the application form with your scratchpad details and hands it back to you.

3. System Architecture & Request Pipeline

Sarkari Sahayak employs an asynchronous, high-throughput pipeline driven by FastAPI and LangGraph. The request-response execution pipeline operates as follows:

When a voice note, text message, or image upload hits the FastAPI webhook, it retrieves or creates the user session. Voice audio is transcribed using Whisper Large-v3, while document uploads are parsed using Llama 4 Scout Vision OCR. The extracted parameters are stored in the session cache.

Next, the LangGraph engine runs a 5-node reasoning cycle to search schemes, filter by rules, suggest complementary packages, and compose the voice/text response. The synthesized speech is returned along with a link to the filled PDF form.

SARKARI SAHAYAK - REQUEST RESPONSE EXECUTION PIPELINE

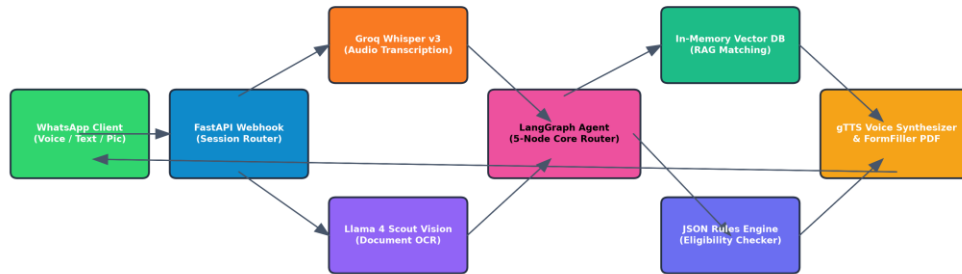


Figure: Diagrammatic representation of System Architecture & Request Pipeline

4. The LangGraph Reasoning Engine Deep-Dive

The core reasoning engine is built as a stateful graph where each node updates a shared state schema:

- Node 1: Extract (Demographics Node): Uses Llama 3.3 to structure demographic details from incoming messages or OCR strings.
- Node 2: Retrieve (RAG Search Node): Queries the database using semantic search and ranks schemes based on cosine similarity of text embeddings.
- Node 3: Evaluate (Rules Node): Iterates over the candidate schemes, comparing rules (age, income caps, land size, gender) against the parsed user profile.
- Node 4: Chain (Forward Chaining Node): Infers supplementary opportunities. If primary schemes are eligible, it automatically appends related welfare options.
- Node 5: Compose (Response Node): Directs Llama 3.3 to write a friendly, conversational synthesis of the eligible programs and next steps.

5. Option A Privacy-First Data Compliance

Handling personal identification records (Aadhaar cards, bank details, income files) demands strict security compliance. Sarkari Sahayak adopts an Option A Privacy architecture:

- Ephemeral State Caching: Extracted details are held temporarily inside memory/Redis caches during the turn. No PII is written to persistent SQL logs.
- Zero Residual Traces: Session caches are destroyed when the user clicks 'Clear Session' or terminates the chat flow.

- Persistent Session Auditing: Only non-PII diagnostic logs (anonymous counts, transaction latency, and model tokens) are permanently saved.

SARKARI SAHAYAK - PRIVACY MODEL (OPTION A COMPLIANCE)

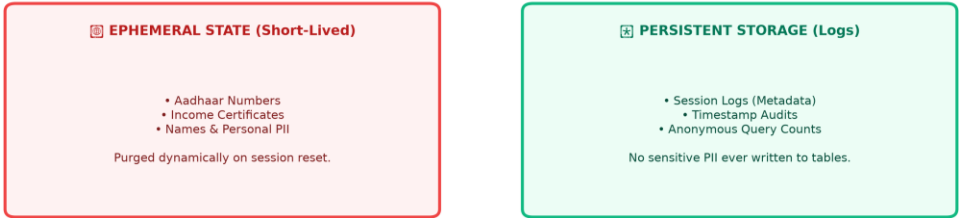


Figure: Diagrammatic representation of Option A Privacy-First Data Compliance

Appendix: Seeded RAG Schemes Registry (50 Schemes)

The database contains 50 active-user schemes loaded from 'schemes_seed.json' for the RAG engine:

Scheme Name	Category	State	Key Description
PM-Kisan Samman Nidhi	Agriculture	Uttar Pradesh	Financial support of ₹6,000 per year for landowning farmers across India.
UP Senior Pension Scheme	Pension	Uttar Pradesh	Old age pension monthly assistance of ₹1,000 for senior citizens in UP.
UP Kanya Sumangala Yojana	Women & Child	Uttar Pradesh	Financial assistance of ₹15,000 for girls' education, health, and welfare in Uttar Pradesh.
UP Free Tablet Smartphone Yojana	Education	Uttar Pradesh	Free tablets and smartphones for college students and youth pursuing higher education in Uttar Pradesh.
Ayushman Bharat Yojana	Health	All India	Universal health insurance coverage providing up to ₹5 Lakhs per family per year for secondary and tertiary

			hospitalization.
Pradhan Mantri Awas Yojana - Gramin	Housing	All India	Provides financial assistance of ₹1.2 Lakhs to rural families for constructing pucca houses with basic amenities.
UP Widow Pension Scheme	Women & Child	Uttar Pradesh	Pension assistance of ₹1,000 per month for widows residing in Uttar Pradesh.
Pradhan Mantri Ujjwala Yojana	Energy & Power	All India	Free LPG connections and cylinder support to women from below poverty line (BPL) households.
UP Divyangjan Pension Yojana	Pension	Uttar Pradesh	Financial monthly assistance of ₹1,000 for disabled persons (40% or more disability) in Uttar Pradesh.
Pradhan Mantri Shram Yogi Maandhan	Pension	All India	Voluntary and contributory pension scheme for unorganized sector workers providing ₹3,000/month after age 60.
UP Shadi Bimari Yojana	Women & Child	Uttar Pradesh	Financial marriage grant of ₹20,000 for daughters of poor families in Uttar Pradesh.
National Social Assistance Programme - IGNDPS	Pension	All India	Disability pension for BPL citizens aged between 18 and 79 with severe or multiple disabilities.
UP Free Cycle Scheme	Labour Support	Uttar Pradesh	Provides free cycles to registered construction laborers in Uttar Pradesh to commute to work.
Pradhan Mantri Mudra Yojana - Shishu	Business Loan	All India	Collateral-free loans up to ₹50,000 for starting small businesses or shops.
Pradhan Mantri Mudra Yojana -	Business Loan	All India	Collateral-free business loans

Kishor			between ₹50,000 and ₹5 Lakhs for expanding micro enterprises.
Pradhan Mantri Mudra Yojana - Tarun	Business Loan	All India	Collateral-free business expansion loans between ₹5 Lakhs and ₹10 Lakhs for established businesses.
Atal Pension Yojana	Pension	All India	Government-backed pension scheme providing monthly pension of ₹1,000 to ₹5,000 after age 60.
Pradhan Mantri Suraksha Bima Yojana	Insurance	All India	Accidental death and disability insurance cover of ₹2 Lakhs at a premium of just ₹20 per year.
Pradhan Mantri Jeevan Jyoti Bima Yojana	Insurance	All India	Life insurance cover of ₹2 Lakhs in case of death due to any reason, at a premium of ₹436 per year.
PM Street Vendor AtmaNirbhar Nidhi	Business Loan	All India	Special micro-credit facility providing working capital loans up to ₹10,000 for street vendors.
PM Vishwakarma Scheme	Labour Support	All India	Skill training, toolkit incentive of ₹15,000, and low-interest collateral-free loans for traditional artisans and craftsmen.
UP Post-Matric Scholarship Scheme	Education	Uttar Pradesh	Tuition fee reimbursement and maintenance stipend for UP students studying in Class 11, 12, or higher courses.
UP Pre-Matric Scholarship Scheme	Education	Uttar Pradesh	Financial scholarship for students studying in Class 9 and 10 in Uttar Pradesh schools.
Mahatma Gandhi National Rural Employment Guarantee Act	Employment	All India	Provides at least 100 days of guaranteed wage employment in a financial year to every

			rural household whose adult members volunteer to do unskilled manual work.
Deen Dayal Upadhyaya Grameen Kaushalya Yojana	Employment	All India	Placement-linked skill training program for rural poor youth to enhance employability and income.
Prime Minister Employment Generation Programme	Business Loan	All India	Credit-linked subsidy program for setting up new micro-enterprises in manufacturing or service sectors, offering subsidies up to 35%.
National Means-cum-Merit Scholarship	Education	All India	Scholarship of ₹12,000 per year awarded to meritorious students from economically weaker sections to prevent drop-outs after Class 8.
Pradhan Mantri Fasal Bima Yojana	Agriculture	All India	Crop insurance scheme offering financial support to farmers suffering crop loss or damage due to natural calamities.
Kisan Credit Card Scheme	Agriculture	All India	Provides farmers with timely credit cards to meet their cultivation and crop production needs at a very low interest rate.
UP Mukhyamantri Abhyuday Yojana	Education	Uttar Pradesh	Free coaching classes (physical and virtual) for competitive exams like UPSC, JEE, NEET, and NDA for students in UP.
UP Bhagirathi River Fishery Scheme	Agriculture	Uttar Pradesh	Financial assistance and subsidies to fishermen and fish breeders for river farming and boat

Pradhan Mantri Matsya Sampada Yojana	Agriculture	All India	purchasing. Flagship scheme for sustainable development of the fisheries sector, supporting aquaculture, deep sea fishing, and cold chain infrastructure.
UP Gopalak Yojana	Agriculture	Uttar Pradesh	Loans up to ₹9 Lakhs for unemployed youth in Uttar Pradesh to establish dairy farms with 10 to 20 cattle.
UP Kamdhenu Dairy Scheme	Agriculture	Uttar Pradesh	Capital subsidy and commercial bank loans for starting high-yield milk-producing dairy centers in rural UP.
UP Solar Pump Yojana	Agriculture	Uttar Pradesh	Provides up to 60% subsidy to farmers for installing solar water pumps for agricultural irrigation.
UP Atal Residential School Admission Scheme	Education	Uttar Pradesh	Free high-quality residential boarding school education (class 6 to 12) for the children of registered laborers.
UP Matritva Shishu Evam Balika Madad Yojana	Labour Support	Uttar Pradesh	Financial assistance of ₹25,000 for maternity costs and newborn child care to registered construction workers.
UP Shramik Awas Sahayata Yojana	Housing	Uttar Pradesh	Financial aid of ₹1 Lakh for house construction or repairs to registered building construction laborers.
UP Mukhyamantri Bal Seva Yojana	Social Welfare	Uttar Pradesh	Financial assistance of ₹4,000 per month for orphans or children who lost breadwinners to pandemics.
Sovereign Gold Bond	Investment	All India	Government

Scheme			securities denominated in grams of gold, offering a secure gold investment with 2.5% annual interest.
Stand-Up India Scheme	Business Loan	All India	Bank loans between ₹10 Lakhs and ₹1 Crore to at least one SC/ST borrower and one woman borrower per bank branch for setting up a greenfield enterprise.
Sukanya Samriddhi Yojana	Investment	All India	A high-interest savings scheme targeted at parents of girl children to build a fund for their higher education and marriage.
UP Sant Ravidas Shiksha Sahayata Yojana	Education	Uttar Pradesh	Monthly education stipend from ₹150 to ₹2,000 for children of registered laborers studying in class 1 to higher degrees.
UP Kaushal Vikas Mission Training Scheme	Employment	Uttar Pradesh	Free skill training and certification in high-demand industrial sectors for unemployed youth in UP.
UP ODOP Toolkit Training Scheme	Labour Support	Uttar Pradesh	Skill upgrade training and free modern toolkits distributed to local artisans under the One District One Product scheme.
PM Programme for Restoration Awareness Generation Nourishment of Mother Earth	Agriculture	All India	Promotes balanced use of chemical fertilizers along with organic and bio-fertilizers, offering subsidies on soil health testing.
UP Free Boring Scheme	Agriculture	Uttar Pradesh	Provides financial grant and boring facilities to small and marginal farmers in

			UP for irrigation tube wells.
UP Mukhyamantri Audyogik Kushta Pension	Pension	Uttar Pradesh	Monthly pension assistance of ₹3,000 for leprosy-affected individuals residing in UP.
National Career Service Portal Registration	Employment	All India	A single-point platform offering job matching, career counseling, and vocational guidance to job seekers across India.
Central Sector Scheme of Scholarship for College and University Students	Education	All India	Scholarship awarded to top 80th percentile board students pursuing graduate or post-graduate courses, providing ₹12,000 per year.